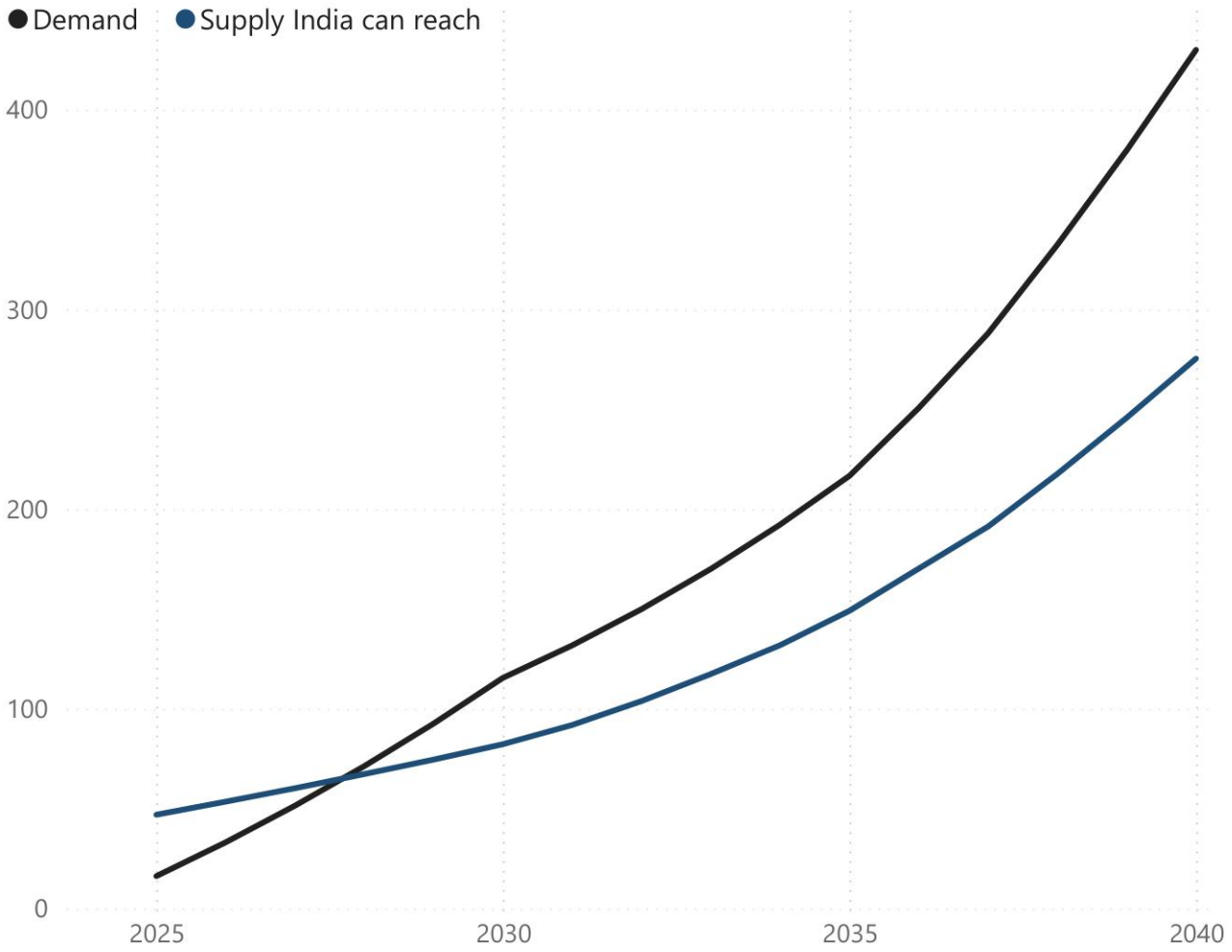
India will need 27 times more lithium by 2040. From 2028, there isn't enough for EVs and storage.

Thousand tonnes of lithium carbonate equivalent (kt LCE) a year. High case. India gets 3% of the world's lithium.



THREE THINGS TO KNOW

38%

of India's lithium in 2025 went to phones, laptops, grease, glass and defence, not clean energy. Non-battery uses alone: 11%, as worldwide (12%).

62%

of EV and storage demand can be met in 2040. Phones and grease are served first because they can pay more.

Overseas assets

is the biggest single fix. It closes 50 of the 155 kt gap in 2040.

Scenario: High

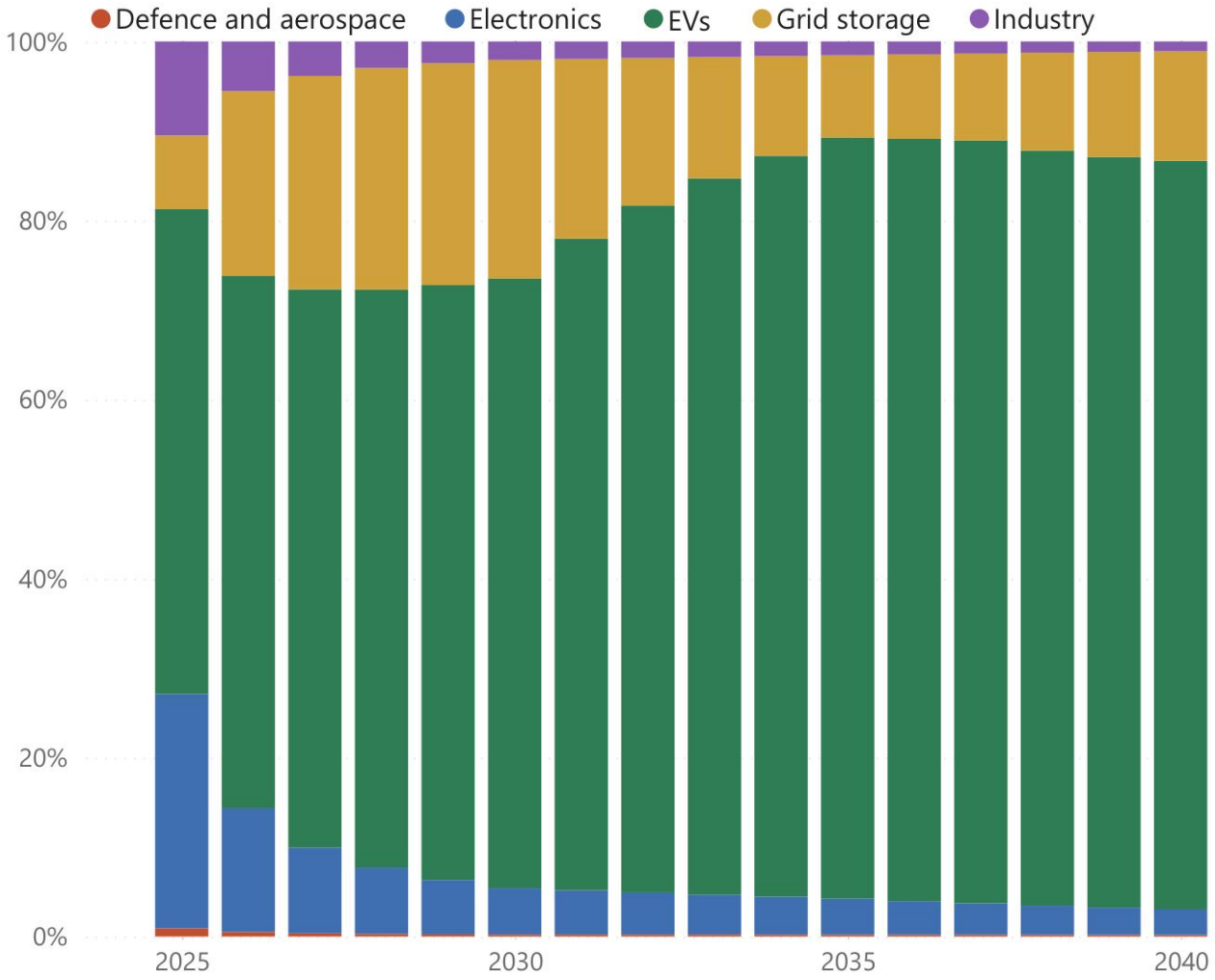
Policy lever: None

World share: 3%

Source: model built on USGS, IEA, CEA, IESA, IDC and UN Comtrade data. "Supply India can reach" = recycling + mining in India + overseas mines + India's share of world output.

In 2025, 38% of India's lithium went to phones, grease and other non-energy uses. By 2040, EVs take 84%.

Share of lithium demand by use, High case.



THREE THINGS TO KNOW

1.2 kt

a year goes into phones assembled in India for export. That lithium leaves the country and never comes back for recycling.

1.1 kt

goes into lubricating grease. 77% of the lithium hydroxide behind it came from Russia in 2023.

11%

goes to grease, glass and medicines, close to the 12% of world lithium used outside batteries (USGS). Most of the rest is phone and laptop batteries.

Scenario: High

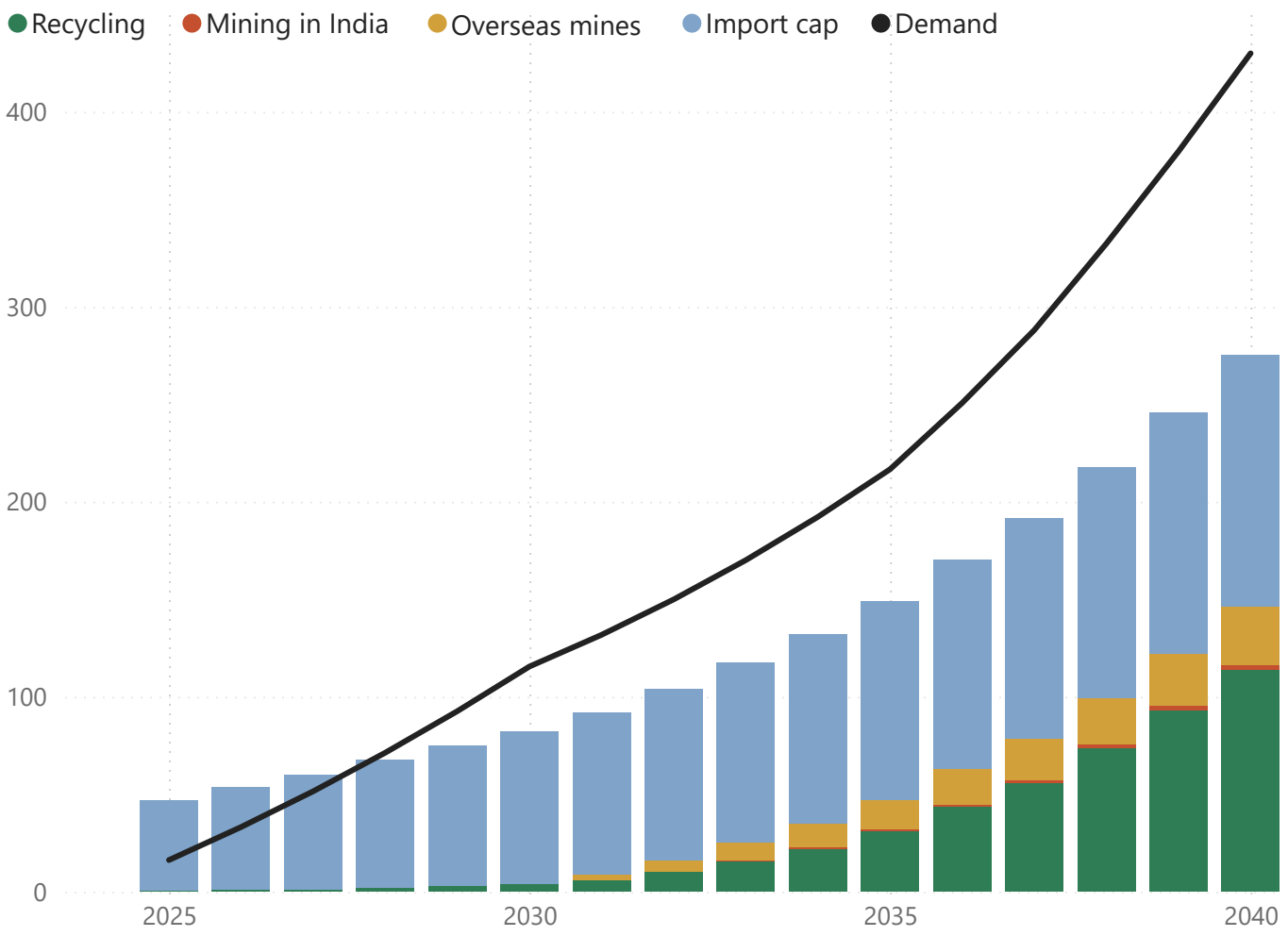
Policy lever: None

World share: 3%

Source: model built on USGS, IEA, CEA, IESA, IDC and UN Comtrade data. Defence is an estimate, since no public data exists.

Recycling becomes India's biggest home source of lithium, but only from the mid-2030s.

Supply India can reach by source, against demand (kt LCE a year). High case, 3% of world supply.



THREE THINGS TO KNOW

4%

is the recycled content India's own scrap can supply in 2030. The battery rules ask for 20%. Most batteries sold today retire after 2032.

18 kt

is all the lithium in Reasi, India's only proven find (583 ppm, Lok Sabha 2024). That is about seven weeks of 2035 demand in the Base case.

73%

of India's lithium is still imported in 2040, down from 96% in 2025.

Scenario: High

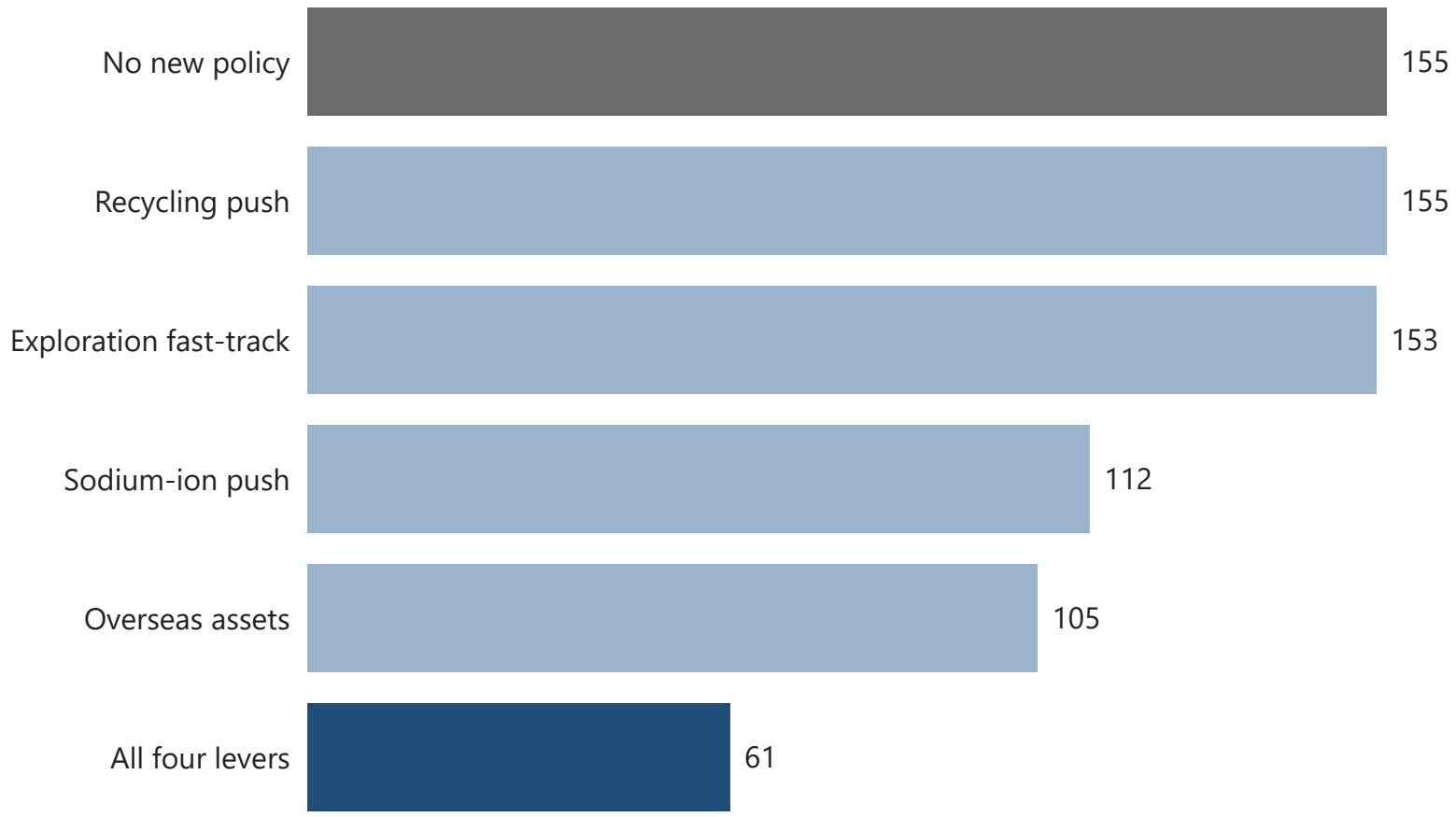
Policy lever: None

World share: 3%

Source: model built on USGS, IEA, CEA, IESA, IDC and UN Comtrade data. Imports are capped at India's share of world output, set with the slicer.

Buying into overseas mines closes more of the gap than any other step. Mining at home closes only 1.5 kt of it.

Lithium gap left in 2040 with each policy lever (kt LCE). High case, 3% of world supply. Gap with no new policy: 155 kt.



INDIA'S VALUE CHAIN TODAY

Stage	Where India is
1 Exploration	Early
2 Mining	None
3 Beneficiation and processing	R&D only
4 Refining (carbonate, hydroxide)	None operating
5 Cathode active material	Negligible
6 Cell manufacturing	Starting
7 Pack assembly and use	Established
8 Recycling	Scaling

India is missing almost every step between the mine and the cell. Lithium is only about $5 of each kWh of battery, so the money is in cathode and cell making, not in mining.

Scenario: High

Year: 2040

World share: 3%

Source: model built on USGS, IEA, CEA, IESA, IDC and UN Comtrade data. Levers: Recycling push, Overseas assets, Sodium-ion push and Exploration fast-track are defined in the model workbook.